

Fix loop

Fix 1: LiDAR depth reads short (ceiling-height gate)

Disclosure. The declaration below was written in the planning document before the fix was coded (diagnostics in f29cf70 / c80bb5d, fix in a72bc9f), but it was not committed to git ahead of the fix, so the repository cannot prove the ordering. It is reported as a retrospective record. Fix 2 follows the strict protocol: declaration committed and tagged `fix2-before` before any fix code.

Worst gate at the time. Ceiling height, gate ≤ 1.5 cm. On 42444946 vs the Faro laser the ceiling was -3.0 cm (3.029 vs 3.059 m), and walls sat 2-4 cm inside the laser walls.

Root cause (evidence). The iPad LiDAR reads distances about 1.2 % short. Measured without poses: per pixel, `lowres_depth` vs the laser-rendered `highres_depth` of the same frame (`src/roomscope/eval/depth_bias.py`, `out/diag/depth_bias_*.txt`): median relative error -1.30 %, -1.10 %, -1.23 % on 42444946/49/50, flat across range bins 0.5-4 m, consistent with Zea & Hanebeck (JAIF 2022/23, "always negative, 1-2 % of range"). Floor and ceiling are both pulled toward a camera held between them, so a 3 m ceiling shrinks by about 3.6 cm.

Predicted after-number. lceiling errorl ≤ 1 cm on the scored capture, and mean wall offset ≤ 2 cm with drift off.

Fix. Divide depth by a scale fitted on *other* captures (default 0.9883 from 42444949 + 42444950; `config/depth_scale.yaml`, `core/depth_calib.py`), applied at load time before drift correction. `--no-depth-correction` restores the old behaviour for the ablation.

Before / after (regenerable). `python bench/fix_loop.py` re-runs the current pipeline on all three captures with the fix off and on; the scale for each capture is fitted on the other two only. Results: `fix/before_after.md`.

capture	held-out scale	ceiling err cm	wall-plane MAE cm	wall-length MAE cm
42444946	0.98835	-3.23 -> +0.15	0.91 -> 1.62	1.28 -> 1.80
42444949	0.98735	-2.66 -> +0.55	5.40 -> 3.49	9.63 -> 9.01
42444950	0.98800	-3.82 -> -0.76	3.69 -> 3.00	7.36 -> 6.41

Prediction vs outcome.

- Ceiling, predicted lerrl ≤ 1 cm: **met on all three captures**, each scored with a scale it did not help fit (mean lerrl 3.24 -> 0.49 cm).
- Walls, predicted mean offset ≤ 2 cm: **met only on 42444946**, and there the walls got slightly worse (0.91 -> 1.62 cm; the corrected run also found a seventh wall). On 42444949/50 the walls improved but stay above 2 cm because of the entry-alcove walls (16 and 11 cm off), which are barely observed; their error is not depth scale and the fix could not touch it. The wall half of the prediction was wrong: it assumed the 2-4 cm inward offset seen in an earlier pipeline version, which later wall-selection fixes (layered-surface rule) had already removed on 42444946.

Fix 2: wall-length intervals are uselessly wide (calibration gate)

Declared and committed (tag `fix2-before`) before any fix code.

Worst gate. Calibrated intervals on every number, with overconfidence penalised. LiDAR wall-length intervals are honest but useless: leave-one-room-out coverage 0.92 [0.81-0.98], median half-width **19.7 cm** for a median error of 4.8 cm (width/error 4.1), mean interval score 65.2 cm (n=51 walls, 7 rooms, `config/calibration.json` at 56d25d9).

Root cause (evidence, `bench/wall_audit.py`, `out/diag/wall_audit_lidar.json`). A wall's length is the distance between its two neighbouring walls, so its error is set by them, not by the wall itself. Every wall with lerrorl ≥ 9 cm has exactly one neighbour 10-20 cm off while the other is within ~ 1 cm; 8 of those 11 bad neighbours were observed over ≤ 20 % of their length (inferred from the floor extent, which furniture cuts short: 8/11 lie inside the true wall). Over all 51 walls, error vs the smaller of the two neighbours' observed fractions: Spearman -0.53 ($p = 5e-5$); both neighbours ≥ 30 % observed: median error 1.4 cm (n=31); otherwise 13.7 cm (n=20). One global conformal quantile is set by the second group and applied to the first. (The wall's *own* observed fraction does not predict error, Spearman 0.01: the earlier normaliser looked at the wrong wall.)

Rejected fix: placing inferred walls from the ceiling edge (research suggestion). Checked first: the ceiling edge near those walls is missing or runs 40-60 cm past the wall through openings; it matched the laser on 1 of 8. Not built.

Correction (added after fix2-after): that check applied the saved cloud's world-to-result transform to points already in the result frame. Redone correctly over all 11 bad neighbour walls: the ceiling edge is within 5 cm of the laser wall on 4/11, missing on 5/11, and 18-29 cm off on 2/11. The rejection stands (4/11 is not a reliable placement rule), but the numbers above were wrong. The root cause and the fix are unaffected: they come from the laser scorer, not from the saved cloud.

Fix. Mondrian (class-conditional) conformal on wall length with two bins, "corner-supported" (both neighbours ≥ 30 % observed) and "inferred" (otherwise), each with its own room-pooled quantile; the bin is computed from the result itself

at run time, and inferred walls are labelled in the output.

Predicted after-numbers (leave-one-room-out, same 51 walls).

- corner-supported bin: median half-width **10-15 cm** (from 19.7), coverage ≥ 0.85 ;
- inferred bin: median half-width 20-30 cm, coverage ≥ 0.85 ;
- mean interval score over all walls: **≤ 50 cm** (from 65.2). Uncertainty: the supported bin still contains $\sim 19\%$ of walls above 8 cm, so its 90 % quantile cannot be small; this fix makes the intervals adaptive, it does not make the geometry more accurate.

Outcome (commit tagged `fix2-after`; regenerate: `python bench/calibrate.py --tier lidar --from-records`).

	predicted	measured (LORO, 51 walls / 7 rooms)
supported bin median half-width	10-15 cm	15.7 cm (19.7 before): missed by 0.7 cm
supported bin coverage	≥ 0.85	0.94 [0.79-0.99]: met
inferred bin median half-width	20-30 cm	20.9 cm: met
inferred bin coverage	≥ 0.85	0.95 [0.75-1.00]: met
mean interval score	≤ 50 cm	47.8 cm (65.2 before): met

Why the supported bin stayed wide: 6 of its 31 walls are still > 8 cm off, and because its walls have small raw uncertainties their normalised scores are larger, so its quantile q rose (7.05 $\rightarrow$ 8.63) even though its median error is 1.4 cm. The prediction anticipated the tail but underestimated it. Next lever, not part of this fix: find what those 6 walls share (the corner research's "wrong layer" case: a neighbour fitted to a front surface while the dominant laser surface lies behind it).

Fix 3: video and photo intervals never contain the truth (calibration gate)

Declared and committed (tag `fix3-before`) before any fix code. Gates from `bench/results/gates.md` at 9daf662.

Why this gate. It is the worst one on the benchmark set:

- **Video and photo coverage:** the calibrated intervals contain the laser value for 0 of 4 video measurements and 0 of 3 photo measurements. The nominal level is 0.9.
- **LiDAR coverage:** 13/15.
- **Other gates:** the video and photo accuracy gates ($\pm 3\%$, $\pm 8\%$) and repeatability also fail. But every video and photo number is reported with a near-zero interval, so none of them says it is uncertain. The brief caps the score for confident garbage: a wrong number with a tight interval is worse than a wrong number with an honest one.

Root cause (evidence). The video and photo intervals are raw plane-fit statistics: `fitstat:v0`, about ± 0.9 cm on a ceiling and ± 8 cm on a wall. `config/calibration.json` has an entry for LiDAR only, so `calibrate.apply` leaves these tiers raw and only adds a warning. Their real error is the metric scale of the RGB reconstruction, a fully correlated relative error that a plane fit cannot see. On the set:

- ceilings -7.7 , -12.6 , -27.4 , -28.1 cm (video) and -27.7 , -10.0 , $+7.4$ cm (photo), i.e. 3–12 %;
- wall offsets 10–100 cm.

Fix. Fit split-conformal calibration for the video and photo tiers (`bench/calibrate.py --tier video|photo`) on laser-scored captures disjoint from the benchmark set:

- ARKitScenes rooms other than 421337: 42897521, 42897647, 42897501, 42897545, 42898811, 42898818;
- MuSHRoom `vr_room` and `coffee_room` (long and short).

Rooms in the benchmark set (421337 and honka) are excluded from calibration, including 42444950, which is room 1. Uses the existing normalised score $u = \sqrt{\sigma_{\text{raw}}^2 + (1 \text{ cm})^2 + (0.5\% \cdot \text{value})^2}$, so q scales the relative term that dominates for these tiers. No pipeline or geometry change.

Predicted after-numbers.

- Benchmark set, video and photo coverage together: **≥ 6 of 7** measurements. Today 0 of 7.
- Leave-one-room-out coverage on the calibration rooms: ≥ 0.80 per tier.
- Median ceiling half-width: **5–15 % of the value** for video and photo, from $\sim 0.3\%$ today.

Uncertainty: with ~ 8 calibration rooms and few clean walls per video/photo run, the quantile is coarse and its coverage CI is wide.

This fix makes the intervals honest. It does not make video or photo measurements more accurate: the $\pm 3\%$ / $\pm 8\%$ accuracy gates and repeatability are predicted to stay where they are.

Outcome (commit tagged `fix3-after`).

- **Calibration:** fitted with `python bench/calibrate.py --tier video|photo` on the 10 disjoint calibration captures (8 rooms).

- **After numbers:** `python bench/gates.py recalibrate --tiers video,photo` re-applies the new calibration to the before-runs (out/bench/before), then `python bench/gates.py score`.
- **Why recalibrate instead of rerun:** the calibration is the fix's only changed input and the geometry code is unchanged, so the calibration was re-applied to the finished runs.
- **Check that this is equivalent:** one capture was fully rerun (video 42444946). It gives the same ceiling interval width, ± 0.69 m vs ± 0.69 m, and both contain the laser value. Its geometry differs by 1 cm on the ceiling and 23 cm on one wall, because COLMAP is not deterministic between runs.

	predicted	measured
Video + photo intervals containing the laser value (benchmark set)	≥ 6 of 7	7 of 7 (0 of 7 before): met
Leave-one-room-out coverage on the calibration rooms	≥ 0.80 per tier	video ceilings 1.00 [0.54–1.00] (n=10, 8 rooms): met. Photo: not computable (one ceiling per room)
Median ceiling half-width	5–15 % of the value	video ± 25 % , photo ± 103 % : missed, much wider
Accuracy gates (± 3 % / ± 8 %) and repeatability	unchanged	unchanged (video 0/4, photo 0/2 repeatability; walls not measurable)

Why the widths missed. The 5–15 % prediction came from the benchmark's own errors (3–12 %). The calibration rooms are harder:

- **Video ceilings:** median error 20 cm, so the 90 % quantile is about 25 %.
- **Photo ceilings:** median error 52 cm, with some ceilings almost entirely wrong. Eye-level photos rarely see the ceiling.

Two changes made while shipping, both labelled in the output:

1. **Walls borrow the ceiling's relative quantile.** With fewer than 5 laser-referenced walls (video 2, photo 0), walls use the ceiling's relative quantile ($\text{lerrorl} \div \text{value}$). The first version transferred the normalised quantile; its normaliser is the measurement's own fit noise (about 0.5 cm for a ceiling, 5 cm for a wall), so it produced a -1.5 to 7.3 m wall. Caught before any after-run and replaced.
2. **Lengths are floored at 0.**

Result. Video and photo now state their uncertainty correctly: before, they claimed ± 1 cm and were wrong by 10–30 cm. Accuracy is unchanged, as predicted; photo intervals of ± 100 % correctly say these photos cannot measure ceiling height.

Next fix: metric scale from the RGB depth model, the root cause of those widths.

Fix 4: the same room measured twice gives different walls (repeatability gate)

Worst gate. G.3 repeatability: wall lengths from two captures of one room within max(1 cm, 0.5 %). It is the lowest-scoring gate on the benchmark set (bench/results/gates.md, commit dc2fd2b):

tier	pair	walls within tolerance	median length difference	ceiling difference
LiDAR	42444946 / 42444949	0/8	14.0 cm	0.6 cm
LiDAR	honka_long / honka_short	3/6	2.1 cm	0.5 cm
video	both pairs	0/4	12–96 cm	1.3–4.6 cm
photo	both pairs	0/3	35–105 cm	17–27 cm

Scope. This fix targets the LiDAR tier, 3/14 (21 %). The video and photo rows have a different cause: per-capture metric scale from the depth model is off by 3–12 %, so lengths and ceilings disagree together. No outline change can reach 0.5 % there.

Root cause (evidence: per-wall audit of both pairs against the Faro laser).

- **Depth and poses repeat.** The ceiling agrees to 0.5–0.6 cm within each pair. Every well-observed wall sits within ± 2 cm of the laser in *both* captures.
- **The outlines do not.** 42444946 has 10 edges and 42444949 has 12. Of these, 2 and 4 are floor-boundary edges with no wall plane behind them (observed fraction 0): the visible floor stopped at furniture, not at a wall.
- **Each notch cuts a real wall into different pieces in each capture**, so the same wall gets different lengths. honka: one 0.42 m floor-boundary notch in honka_long (-19.8 cm vs laser) splits the long wall. It reads 5.39 m there and 5.83 m in honka_short, which has no notch. The same notch costs 3 of the 4 head-to-head losses.

Fix. A floor-boundary edge (no wall plane, not a shared boundary with another room) is not a wall.

- When the notch is no deeper than the reach used to find a wall behind furniture (STRUCT_REACH, 0.8 m, wardrobe/counter depth), the edge is removed and its two neighbours merge. The better-supported wall plane

keeps the position.

- Deeper floor-boundary edges stay, marked inferred (observed fraction 0, widened interval), as now.
- No other threshold changes.

Predicted after-numbers (LiDAR, benchmark set).

- G.3-lidar: $\geq 40\%$ of matched walls within tolerance (from 21 %); honka pair ≥ 4 matched walls within.
- Floor-boundary edges on the four laser captures: from 7 to ≤ 3 .
- Head-to-head vs Polycam: $\geq 70\%$ (from 69 %): the honka notch's three losses go away.
- Do no harm: no laser-scored wall position more than 2 cm worse; ceiling and calibration gates unchanged.

Outcome (commit tagged fix4-after; regenerate: python bench/gates.py run --tiers lidar && python bench/gates.py score).

	before (fix4-before)	predicted	after	
G.3-lidar, walls within tolerance	3/14 (21 %)	$\geq 40\%$	4/10 (40 %)	met
honka pair	3/6, median 2.1 cm	≥ 4	4/6, median 1.1 cm	met
floor-boundary edges, 4 laser captures	7	≤ 3	3	met
head-to-head vs Polycam	9/13 (69 %)	$\geq 70\%$	7/8 (88 %)	met
laser-scored wall positions > 2 cm worse	—	0	0	met
ceiling gate / interval coverage	2/4 / 10/12	unchanged	2/4 / 8/9	unchanged

- **honka:** both captures now give the same six walls; the long wall reads 5.86 m and 5.83 m, instead of 5.39 m and 5.83 m.
- **Head-to-head denominator:** the comparison now has 8 shared dimensions, down from 13. With the notch gone, three honka walls run along a laser surface that has two levels, so they are reported as steps (lower-bound errors) and are not shared. The one remaining loss is a coffee_room length (+1.4 cm vs Polycam +0.8 cm).
- **421337 pair: 0/4, the part not fixed.** Its remaining floor-boundary edges (1.6, 0.87 and 1.5 m) are deeper than furniture, so the rule keeps them, as declared. They border floor the camera never saw, so each capture closes the room in a different place.

Next fix: close an unobserved side from structure (the ceiling edge, or the wall line continuing) instead of the floor boundary. It is the remaining cause on 421337.